

RESEARCH PAPER

DATA ANALYTICS AND BUSINESS INTELLIGENCE

Product Line Profitability and Margin Performance Analysis

A Data-Driven Study of Nassau Candy Distributor

Research Focus

Analysis of sales, cost, units, gross profit, product and division contribution, regional performance, and profit-concentration patterns using Nassau Candy Distributor's full order-level transaction dataset.

Abstract

Sales volume is frequently used as a proxy for product success, yet volume alone can obscure the true cost of goods sold and mask underlying margin erosion. This paper presents an exploratory data analysis of 10,194 order-level transactions spanning January 2024 through December 2025, covering 15 products across three divisions (Chocolate, Sugar, and Other) distributed to 59 U.S. states. We compute product- and division-level profitability metrics — gross margin, profit per unit, revenue and profit contribution — and apply Pareto (80/20) analysis to assess profit concentration. We find that the portfolio generated \$141,784 in total sales and \$93,443 in gross profit (65.9% blended margin), that the Chocolate division alone contributes 95.1% of total profit, and that just four of fifteen products (26.7%) account for 80% of total profit. Two products — Kazookles and Lickable Wallpaper — are identified as high-risk, combining above-median sales with below-median margin. We conclude with pricing, sourcing, and portfolio-rationalization recommendations.

1. Introduction

For distributors operating across a diverse product catalog, sales volume alone is a misleading indicator of business health. Some products sell in high volume while generating disproportionately low profit, consuming manufacturing or operational cost that erodes overall margin even as top-line revenue appears strong. Nassau Candy Distributor requires visibility into which product lines deliver the highest gross margin, whether high-sales products are actually profitable, how profitability varies across product divisions, and which products represent ongoing margin risk. Without this insight, decisions on pricing, promotions, and product portfolio composition remain reactive and intuition-based rather than evidence-based.

This paper addresses that gap by transforming raw order-level transaction data into product-, division-, and portfolio-level profitability intelligence, using standard exploratory data analysis (EDA) techniques together with profit-concentration (Pareto) analysis.

1.1 Analytical Framing: Margin Analysis and the Pareto Principle

Two established analytical concepts anchor this study. Gross margin analysis — expressing profit as a percentage of sales rather than an absolute figure — allows fair comparison across products and divisions of very different scale, and is the standard first step in retail and distribution profitability review. The Pareto principle, commonly summarized as the '80/20 rule,' observes that in most business portfolios a small share of items account for the majority of an outcome of interest (here, profit). Applying Pareto analysis to a product catalog surfaces both the items that most warrant protection and investment, and the concentration risk that comes with depending on a narrow subset of the catalog. Both techniques are used together in this paper to move beyond a single top-line profitability figure toward an actionable, product-level view.

2. Data and Methodology

2.1 Dataset

The dataset comprises 10,194 valid order-line records (after removing zero-sales and invalid-profit rows) drawn from 8,549 unique orders placed by 5,044 unique customers between January 2, 2024 and December 31, 2025. Each record includes order and shipment metadata (order date, ship date, ship mode), customer geography (country, city, state, postal code, region), product identifiers (division, product ID, product name), and financial fields (sales, units, cost, and gross profit, defined as sales minus cost).

2.2 Data Cleaning and Validation

- Validated cost and sales values as positive, non-null numeric fields
- Removed zero-sales or invalid-profit records
- Standardized product and division labels for consistent grouping
- Parsed order dates for time-series aggregation

2.3 Metric Definitions

Metric	Definition
Gross Margin (%)	Gross Profit ÷ Sales
Profit per Unit	Gross Profit ÷ Units
Revenue Contribution	Product Sales ÷ Total Sales
Profit Contribution	Product Profit ÷ Total Profit
Margin Volatility	Standard deviation of margin (%) across a product's order lines

Table 1. Key performance indicators used throughout this analysis.

3. Exploratory Data Analysis

3.1 Portfolio Overview

The portfolio spans three divisions — Chocolate, Sugar, and Other — comprising 15 distinct products. Order-level gross margin ranges from 7.7% to 80.0%, with a mean of 66.5% and a standard deviation of 6.7 percentage points, indicating that most products are priced within a relatively narrow, consistent margin band, with a small number of clear outliers.

3.2 Cost–Sales Relationship

Cost and sales are strongly positively correlated ($r = 0.96$) at the order-line level, and unit volume correlates with gross profit at $r = 0.82$. This confirms that, for most of the catalog, unit economics are stable and predictable — profit scales with volume — which makes the exceptions to this pattern (see Section 4.3) particularly important to isolate. Figure 2 visualizes this relationship, coloring each order line by division; the tight, near-linear clustering confirms consistent per-unit pricing, while any point falling well off the main diagonal would indicate an anomalous cost or pricing event.

Figure 2. Cost vs. Sales by Division (bubble size = gross profit)

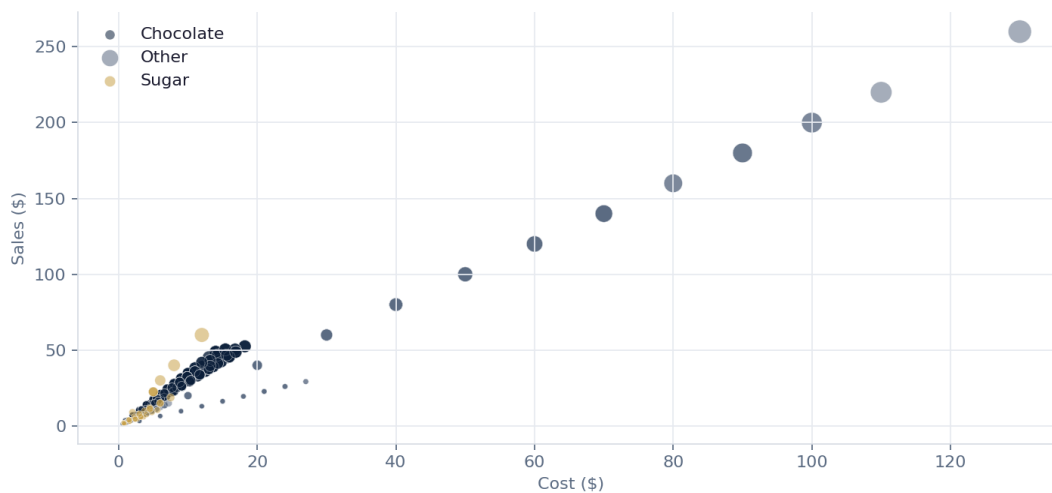


Figure 2. Cost vs. Sales by division; bubble size denotes gross profit per order line.

3.3 Margin Stability by Product

Within-product margin volatility is effectively zero for every product in the catalog (standard deviation of margin across a product's own order lines is near 0.0 in all 15 cases). This indicates that Nassau Candy applies consistent, fixed per-unit pricing and costing for each SKU — variation in blended margin at the division or portfolio level is therefore driven entirely by product mix, not by inconsistent pricing execution.

3.4 Temporal Trend

Aggregating sales by month across the 24-month observation window shows a clear growth trajectory: monthly sales rose from roughly \$2,100 in January 2024 to roughly \$12,300 in December 2025, with monthly gross profit following the same upward trend (Figure 4). This suggests the profitability findings in this paper reflect an expanding, not static, business, and should be revisited periodically as new order data accumulates.

Figure 4. Monthly Sales and Gross Profit Trend (Jan 2024 - Dec 2025)

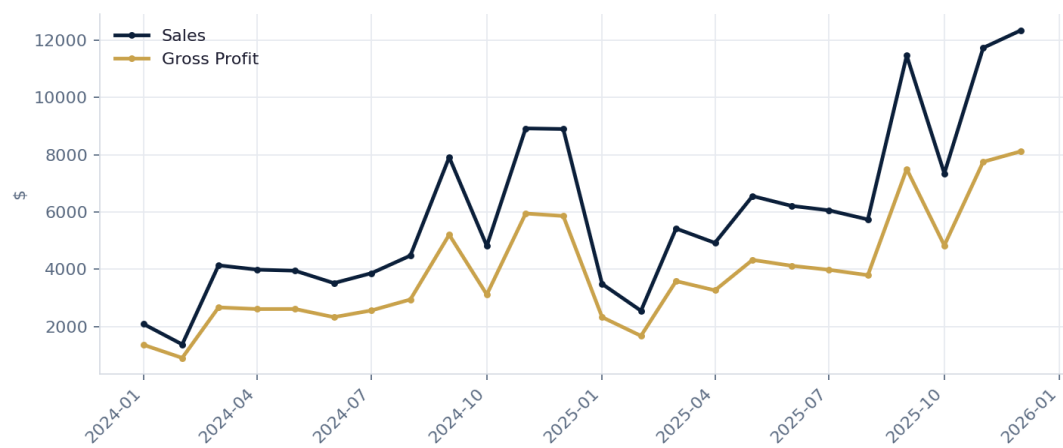


Figure 4. Monthly sales and gross profit, January 2024 – December 2025.

3.5 Geographic Distribution

Orders originate from 59 U.S. states, grouped into four regions — Pacific, Atlantic, Interior, and Gulf. The Pacific region leads in both sales (\$46,302) and profit (\$30,486), followed by Atlantic (\$41,197 sales / \$26,974 profit). Margin is remarkably consistent across regions (65.5%–66.4%), indicating that regional differences are driven by demand volume rather than regional pricing or cost disparities. California, New York, and Texas are the top three states by sales.

Region	Sales	Gross Profit	Margin
Pacific	\$46,301.53	\$30,485.94	65.8%
Atlantic	\$41,197.24	\$26,973.70	65.5%
Interior	\$32,037.60	\$21,282.49	66.4%
Gulf	\$22,247.26	\$14,700.67	66.1%

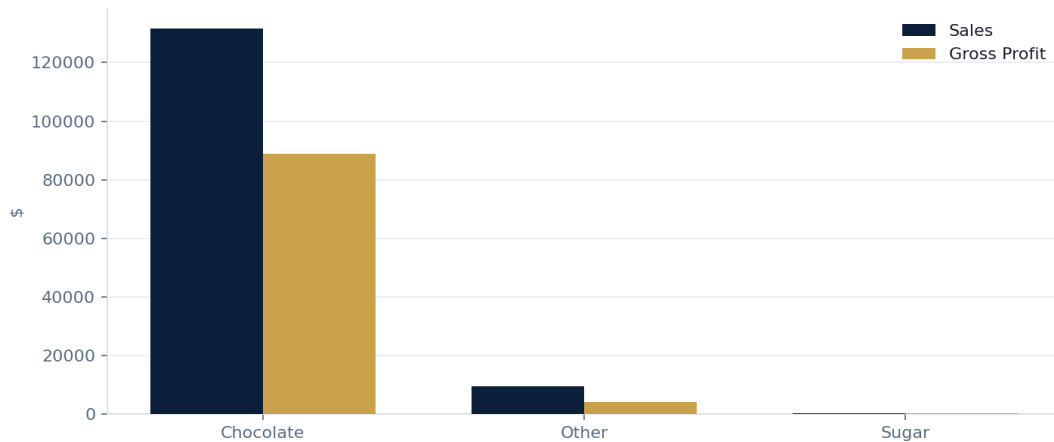
Table 2. Regional performance summary.

4. Findings

4.1 Division-Level Profitability

Division	Sales	Gross Profit	Gross Margin	Share of Profit
Chocolate	\$131,692.90	\$88,824.62	67.4%	95.1%
Other	\$9,663.25	\$4,333.45	44.8%	4.6%
Sugar	\$427.48	\$284.73	66.6%	0.3%

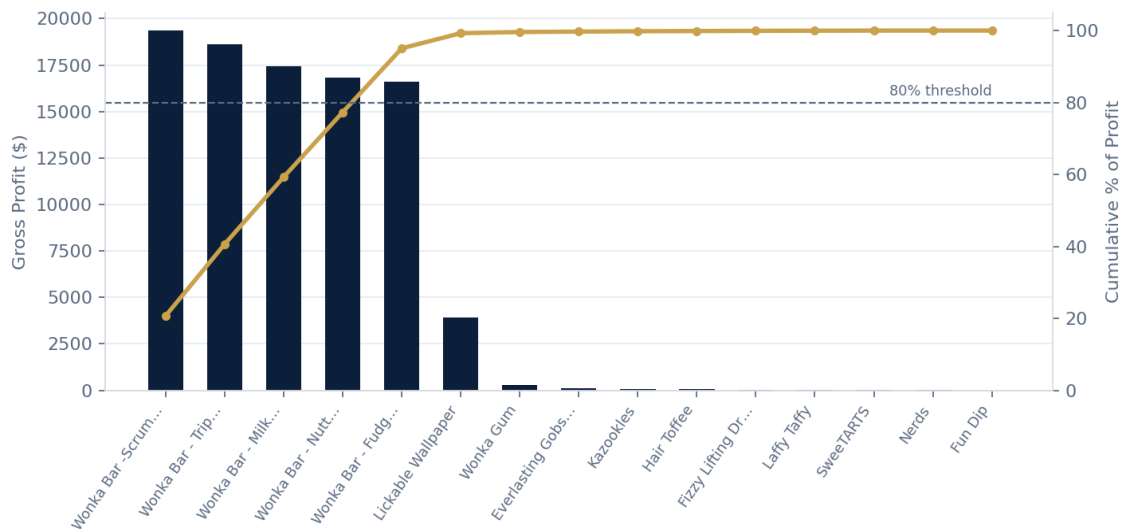
Table 3. Division performance summary.

Figure 3. Sales vs. Gross Profit by Division*Figure 3. Sales vs. gross profit by division.*

The Chocolate division is the overwhelming driver of business performance, contributing 92.9% of total sales and 95.1% of total gross profit at a healthy 67.4% margin. The Other division, while a minor contributor by volume (6.8% of sales), carries a materially weaker 44.8% margin — 22.6 percentage points below Chocolate — flagging it as a candidate for cost or pricing review. The Sugar division is negligible in absolute terms but maintains a healthy margin comparable to Chocolate.

4.2 Profit Concentration (Pareto Analysis)

Ranking all 15 products by gross profit and computing cumulative contribution shows that just 4 products (26.7% of the catalog) account for 80% of total portfolio profit (Figure 1). This is a textbook Pareto pattern and signals a meaningfully concentrated — and therefore fragile — profit base: the loss or underperformance of any one of these four products would have an outsized effect on total profitability.

Figure 1. Pareto Chart — Profit Concentration by Product*Figure 1. Pareto chart of gross profit by product, with cumulative contribution.*

Rank	Product	Gross Profit	Margin
1	Wonka Bar - Scrumdiddlyumptious	\$19,357.50	69.4%
2	Wonka Bar - Triple Dazzle Caramel	\$18,610.20	65.3%
3	Wonka Bar - Milk Chocolate	\$17,443.37	64.9%

Rank	Product	Gross Profit	Margin
4	Wonka Bar - Nutty Crunch Surprise	\$16,819.95	71.3%

Table 4. The four products responsible for 80% of total gross profit.

4.3 High-Sales, Low-Margin Risk Products

Filtering for products with above-median sales but below-median margin isolates two items that appear successful by volume but underperform financially: Lickable Wallpaper (\$7,860 in sales at a 50.0% margin) and Kazookles (\$1,206 in sales at just 7.7% margin — the single lowest margin in the entire catalog). Kazookles in particular warrants immediate attention, as its margin is less than one-eighth of the portfolio average.

Product	Sales	Margin	Profit per Unit
Lickable Wallpaper	\$7,860.00	50.0%	\$10.00
Kazookles	\$1,205.75	7.7%	\$0.25

Table 5. Products flagged as margin risks.

5. Discussion

Three patterns emerge from this analysis. First, profitability at Nassau Candy is a function of product mix, not pricing inconsistency — margins are stable within each SKU, so portfolio-level margin management is fundamentally a question of which products to grow, sustain, or exit, rather than a question of repricing execution. Second, the business is highly dependent on a narrow set of Chocolate-division products; while this concentration currently correlates with strong margins, it also represents dependency risk should input costs, demand, or supply for those specific items shift. Third, the Other division's comparatively weak 44.8% margin, combined with Kazookles' outlier 7.7% margin, suggests that non-Chocolate categories may be underpriced relative to their cost structure, or sourced less efficiently than the core Chocolate line.

6. Recommendations

- Prioritize continued investment, shelf space, and marketing support for the four core Chocolate products that generate 80% of portfolio profit.
- Conduct an immediate cost and pricing review of Kazookles (7.7% margin); repricing, supplier renegotiation, or discontinuation should be evaluated.
- Investigate the structural cost or pricing drivers behind the Other division's 44.8% margin relative to Chocolate's 67.4%.
- Reduce profit concentration risk over time by developing margin-healthy alternatives outside the current top-four dependency set.
- Continue monitoring the Pacific and Atlantic regions, which together drive the majority of sales, for capacity and fulfillment planning as monthly volume continues its upward trend.
- Operationalize these metrics via the accompanying Streamlit dashboard for continuous, self-service monitoring rather than periodic manual analysis.

7. Limitations and Future Work

This analysis relies on gross profit (sales minus cost of goods) and does not incorporate fixed overhead, logistics, or marketing spend, so true net profitability by product may differ from the gross-margin figures reported here. A fully loaded cost-allocation model — apportioning warehousing, freight, and administrative cost to each product — would sharpen the risk assessment in Section 4.3 and may reveal additional products whose net profitability is weaker than gross margin alone suggests. The dataset also reflects a 15-product catalog; findings on division-level concentration may shift as the product

line expands. Future work should extend this analysis to incorporate fully loaded cost accounting, customer-segment profitability, and seasonal demand forecasting to support proactive, rather than retrospective, portfolio decisions.

8. Conclusion

This exploratory analysis establishes a clear, data-driven view of product line profitability at Nassau Candy Distributor. The Chocolate division, and specifically four flagship products within it, drive the substantial majority of company profit, while a small number of underperforming items — most notably Kazookles — represent concrete, addressable opportunities for margin improvement. Replacing volume-based intuition with margin-based evidence provides Nassau Candy's leadership and government stakeholders with a transparent, auditable foundation for pricing, sourcing, and product-portfolio decisions going forward.

Appendix A: Full Product-Level Performance Table

Division	Product	Sales	Gross Profit	Units	Margin
Chocolate	Wonka Bar -Scrumdiddlyumptious	\$27,874.80	\$19,357.50	7,743	69.4%
Chocolate	Wonka Bar - Triple Dazzle Caramel	\$28,485.00	\$18,610.20	7,596	65.3%
Chocolate	Wonka Bar - Milk Chocolate	\$26,867.75	\$17,443.37	8,267	64.9%
Chocolate	Wonka Bar - Nutty Crunch Surprise	\$23,574.95	\$16,819.95	6,755	71.3%
Chocolate	Wonka Bar - Fudge Mallows	\$24,890.40	\$16,593.60	6,914	66.7%
Other	Lickable Wallpaper	\$7,860.00	\$3,930.00	393	50.0%
Other	Wonka Gum	\$597.50	\$310.70	478	52.0%
Other	Kazookles	\$1,205.75	\$92.75	371	7.7%
Sugar	Everlasting Gobstopper	\$130.00	\$104.00	13	80.0%
Sugar	Hair Toffee	\$76.50	\$59.50	17	77.8%
Sugar	Fizzy Lifting Drinks	\$78.75	\$47.25	21	60.0%
Sugar	Laffy Taffy	\$53.73	\$33.48	27	62.3%
Sugar	SweeTARTS	\$61.50	\$28.70	41	46.7%
Sugar	Nerds	\$15.00	\$7.00	10	46.7%
Sugar	Fun Dip	\$12.00	\$4.80	8	40.0%

Table A1. Complete performance summary for all 15 products, ranked by division then gross profit.